

Intervention of human breast cell carcinogenesis chronically induced by 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine

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More than 85% of breast cancers are sporadic and attributable to long-term exposure to environmental carcinogens, such as those in the diet, through a multistep disease process progressing from non-cancerous to premalignant and malignant stages. The chemical carcinogen 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) is one of the most abundant heterocyclic amines found in high-temperature cooked meats and is recognized as a mammary carcinogen. However, the PhIP's mechanism of action in breast cell carcinogenesis is not clear. Here, we demonstrated, for the first time, that cumulative exposures to PhIP at physiologically achievable, pico to nanomolar concentrations effectively induced progressive carcinogenesis of human breast epithelial MCF10A cells from a non-cancerous stage to premalignant and malignant stages in a dose- and exposure-dependent manner. Progressive carcinogenesis was measured by increasingly-acquired cancer-associated properties of reduced dependence on growth factors, anchorage-independent growth, acinar-conformational disruption, proliferation, migration, invasion, tumorigenicity with metastasis and increased stem-like cell populations. These biological changes were accompanied by biochemical and molecular changes, including upregulated H-Ras gene expression, extracellular signal-regulated kinase (ERK) pathway activation, Nox-1 expression, reactive oxygen species (ROS) elevation, increased HIF-1 α , Sp1, tumor necrosis factor- α , matrix metalloproteinase (MMP)-2, MMP-9, aldehyde dehydrogenase activity and reduced E-cadherin. The Ras-ERK-Nox-ROS pathway played an important role in not only initiation but also maintenance of cellular carcinogenesis induced by PhIP. Using biological, biochemical and molecular changes as targeted endpoints, we identified that the green tea catechin components epicatechin-3-gallate and epigallocatechin-3-gallate, at non-cytotoxic doses, were capable of suppressing PhIP-induced cellular carcinogenesis and tumorigenicity.